

ECON8037: Financial Economics

Week 8 Lecture - The Economics of Cryptocurrencies II: Optimal Design, Welfare, and Payments (Chiu & Koeppl, 2017)

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Where we left off

Lecture I built a Lagos–Wright model with a cryptocurrency:

- Mining is competitive and dissipates all rewards: $MQ = \beta R$.
- Buyers can attack via secret mining. A contract (d, N) is *double-spending proof* iff

$$d < R(N + 1)N.$$

- Settlement cannot be both immediate and final.
- Mining reward is endogenous:

$$R = \frac{Z(\mu - 1) + D\tau}{\bar{N} + 1}.$$

Today. Optimal (μ, τ) , calibration to Bitcoin, welfare, comparison with Fedwire and US debit cards.

Roadmap for Lecture II

- ① Optimal cryptocurrency design (the planner's Ramsey problem).
- ② Calibrating the model to Bitcoin.
- ③ Welfare cost: Bitcoin vs. cash vs. optimal cryptocurrency.
- ④ Best use cases: Fedwire (large-value) vs. US debit cards (retail).
- ⑤ Scalability and policy implications.

Two design instruments

The cryptocurrency designer chooses (μ, τ) :

- $\mu \geq 1$ – gross money growth rate (*seigniorage*).
- $\tau \geq 0$ – transaction fee (fraction of d paid to miners).

Both finance mining rewards (which dissipate as mining cost):

$$\frac{\beta}{\mu} R(\bar{N} + 1) = B(\mu - 1 + \sigma\tau) \int \frac{x(\varepsilon)}{1 - \tau} dF_{\varepsilon}(\varepsilon).$$

Trade-off. More mining $\Rightarrow$ tighter DS-proof constraint $\Rightarrow$ larger trade. But mining is a deadweight loss and inflation/fees distort consumption.

Social welfare

Welfare = average utility per period:

$$\mathcal{W} = B\sigma \int_0^\infty [\delta^{N(\varepsilon)} \varepsilon u(x(\varepsilon)) - x(\varepsilon)] dF_\varepsilon(\varepsilon) - C(\bar{N} + 1). \quad (37)$$

By Lemma 1 (mining dissipation), the mining cost equals $\beta R(\bar{N} + 1)/\mu$, the present value of total rewards.

The planner's problem

$$\max_{\mu, \tau} B \int \sigma \delta^{N(\varepsilon)} \varepsilon u[x(\varepsilon)] - x(\varepsilon) dF_{\varepsilon}(\varepsilon) - \frac{\beta}{\mu} R(\bar{N} + 1) \quad (38)$$

subject to

- $(d(\varepsilon), x(\varepsilon), N(\varepsilon))$ is the buyer's optimal contract given (μ, τ) ;
- DS constraint: $d(\varepsilon) = \frac{\mu}{\beta} \frac{x(\varepsilon)}{1-\tau} \leq R(N(\varepsilon) + 1)N(\varepsilon)$;
- Planner's budget: $\frac{\beta}{\mu} R(\bar{N} + 1) = B(\mu - 1 + \sigma\tau) \int \frac{x(\varepsilon)}{1-\tau} dF_{\varepsilon}(\varepsilon)$.

This is a *Ramsey problem*: planner internalises that agents respond to (μ, τ) .

The optimal design result

Proposition (Optimal design)

The optimal reward structure sets transaction fees to zero and relies only on seigniorage:

$$\tau = 0 \quad \text{and} \quad \mu > 1.$$

Why? Both μ and τ distort consumption, but their incidence differs:

- Inflation tax is paid by *all* balance holders.
- Transaction fees are paid only by buyers who *actually trade* – precisely the agents with the highest valuation of balances.

Levying revenue through inflation smooths the distortion across all buyers.

Intuition for the trade-off

- Without rewards: no mining, no security, no equilibrium.
- With too many rewards: mining is excessive; large deadweight loss.
- Optimum balances the marginal benefit of relaxing the DS constraint against the marginal cost of mining.

The result aligns with the broader monetary literature: *ex ante* taxation (inflation on all balances) dominates *ex post* user fees when the latter target the most elastic margin.

Calibrating the model to Bitcoin

Utility $u(x) = \log(x + b) - \log b$ with $b \approx 0$. A period = 1 day, a trading session = 10 minutes (the average block time).

- $\beta = 0.9999$ (≈ 0.97 annual);
- 2015 Bitcoin supply: 14,342,503 BTC;
- daily money growth: $\mu = (1 + 25/14342503)^{6 \times 24} = 1.00025$, $\approx 9.6\%$ annual;
- $\sigma = 0.0178$ (fraction of BTC spent per day);
- $\tau = 0.0088\%$ (from data on fees);
- $B = 6,873,428$ (max number of average-sized transactions).

Benchmark parameters

Parameter	Value	Target
β	0.9999166	period = 1 day
δ	0.9999994	$\delta = \beta^{1/(1+\tilde{N})}$
μ	1.00025	Bitcoin money growth
τ	0.000088	Bitcoin transaction fee
B	6,873,428	max no. of average-sized transactions
σ	0.0178	velocity per block
α	1	normalisation

Preference shock distribution $F(\varepsilon)$ disciplined by the empirical distribution of transaction sizes in [Ron and Shamir \(2013\)](#).

Effects of money growth

Numerical experiment around the calibrated μ :

- Higher $\mu \Rightarrow$ higher $R \Rightarrow$ higher mining effort, shorter average confirmation lag N .
- Aggregate trade and utility *rise* (DS constraint relaxes, lags shrink).
- Mining cost *rises* linearly with μ .
- Net welfare: **hump-shaped** in μ .

Bitcoin's calibrated μ is on the *high* side: more inflation than is optimal.

Welfare measures

Two consumption-equivalent measures:

- **Welfare cost.** Fraction of consumption a person would give up to use cash under the Friedman rule (zero inflation) instead of cryptocurrency.
- **Equivalent inflation tax.** Cash inflation rate that makes a person indifferent between traditional cash and the cryptocurrency.

Bitcoin is very inefficient

	Bitcoin (benchmark)	Bitcoin (optimal)
$\mu - 1$	9.5%	0.17%
τ	0.0088%	0%
welfare loss	1.41%	0.08%
mining cost (per year)	\$359.98 mn	\$6.90 mn
equivalent inflation tax	230.44%	27.51%

- Bitcoin's welfare cost: $\approx 1.4\%$ of consumption $\approx 500\times$ a low-inflation cash system (whose welfare cost is $\approx 0.003\%$).
- Equivalent inflation tax of **230%**: people would prefer cash even at hyperinflationary rates.

Why is Bitcoin so inefficient?

The driver is the *mining cost* – about \$360 million per year at the calibrated parameters.

- Bitcoin's reward structure (high μ + positive τ) overcompensates miners.
- Optimal design (Proposition: $\tau = 0$, lower μ) reduces mining to \$6.9 million per year.
- Welfare loss falls from 1.41% to **0.08%** – comparable to a cash system with moderate inflation.

The lesson: Bitcoin's inefficiency is largely a *design* problem, not an intrinsic feature of cryptocurrencies.

Retail vs. large value payments

Recalibrate the model to two real-world payment systems (2014 US):

- **US Debit cards** (retail): avg transaction \$38.29; annual volume 59,539 mn.
- **Fedwire** (large value): avg transaction \$6.55 million; annual volume 135 mn.

Period = 30 mins; block length = 1 min. Choose σ and ε to match each system.

Welfare comparison

	Retail (Debit)	Large value (Fedwire)
avg transaction size	\$38.29	\$6,552,236
annual volume	59,539 mn	135 mn
optimal $\mu - 1$	0.038%	0.53%
optimal τ	0%	0%
confirmation lag	2 mins	12 mins
welfare loss	0.00052%	0.0060%
mining cost (per year)	\$4.33 mn	\$22.10 bn
equivalent fee (per transfer)	\$0.0002	\$392.56

- Optimal μ is much lower than Bitcoin's for both systems.
- Retail welfare loss is two orders of magnitude smaller than large-value.

Why is retail more efficient?

- Double-spending incentives *scale with transaction size d* , while mining cost is a public good financed by aggregate volume.
- Small individual transactions $\Rightarrow$ DS constraint is easy to satisfy with little mining.
- Large individual transactions $\Rightarrow$ require lots of mining and long lags.

Equivalent fee comparison:

- Cryptocurrency retail: 0.02 cents/transfer vs. debit card interchange ≈ 23 cents.
- Cryptocurrency large value: \$392 vs. Fedwire service fee ≈ 82 cents.

Cryptocurrencies can plausibly compete with retail systems; for large-value transactions, centralised systems remain far more efficient.

Scalability is the binding constraint

For cryptocurrencies to compete with debit cards they must process *enormous* transaction volumes.

- Bitcoin: ≈ 7 transactions per second (TPS) cap.
- Visa: $\approx 1,700$ TPS sustained, peaks $> 24,000$.
- Block-size limits, network latency, propagation delays all bind.

The model takes scaling as exogenous, but the welfare results say: *if technological scalability is solved, optimally designed cryptocurrencies can challenge retail payment networks.*

Caveats

The comparison is suggestive, not definitive:

- Ignores private costs: data storage, wallets, exchanges, software.
- Mining cost is a deadweight loss; some of Visa/Fedwire's fees are provider *profits*, not operating cost.
- Ignores network/scaling constraints (assumed solved).
- Ignores volatility of the cryptocurrency's value.

Policy implications

- **Central bank digital currency (CBDC).** The model puts bounds on the cost of issuing digital currency centrally vs. via PoW.
- **Reward design.** Existing cryptocurrencies should consider eliminating transaction fees and using only seigniorage.
- **Reform Bitcoin?** Lower money growth and remove fees $\Rightarrow$ welfare cost drops by an order of magnitude.
- **Scaling debate.** On-chain scaling is the binding technological constraint if cryptocurrencies are to displace cards.

What we've learned

- 1 **Optimal design.** Finance mining via seigniorage only ($\tau = 0$, $\mu > 1$). Inflation spreads the distortion across all balance holders.
- 2 **Bitcoin is poorly designed.** Welfare cost 1.4% of consumption, mostly wasted on mining.
- 3 **Best use case is retail.** Cryptocurrencies scale efficiently in volume but not in transaction size.
- 4 **Scaling is the real bottleneck.** If technological limits can be overcome, well-designed cryptocurrencies could compete with debit cards.

Open questions and extensions

- Endogenous scalability: model block size, latency, network propagation.
- Alternative consensus protocols (proof-of-stake, BFT).
- Volatility and price discovery for the cryptocurrency.
- Interaction with central bank policy (CBDCs, negative rates).
- Coordinated double-spending attacks (footnote 21: $dn_b < R(N + 1)N$).
- Smart contracts and programmable money.

Thank you

References I

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